End-semester exam timetabling

Batch 7 - 23I501 Design and Analysis of Algorithms

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Technique: **Backtracking** (forward checking + most-constrained-first)

Members: _____

1. Problem modelling

Graph G=(V,E): V = courses, edge = at least one common student. Timetable = slot(v) in {1..k}, slot(u) != slot(v) for every edge.

Real size: n = 420, 9 days x 2 slots. Instance: 7 courses, 3 slots, 10 conflicts.

Question	Type
Can it fit in 3 slots?	Decision (3-colouring, NP-complete)
Minimum number of slots?	Optimisation (chromatic number, NP-hard)
List all valid timetables	Enumeration (chromatic polynomial P(G,3) = 36 labelled)

2. Data structure analysis

Operation	Adjacency sets (chosen)	Adjacency matrix	Edge list
Space	O(n+m)	O(n^2)	O(m)
Conflict test u,v	O(1) avg	O(1)	O(m)
Neighbours of v	O(deg v)	O(n)	O(m)

Domains as bitmasks (k <= 18 bits): remove slot O(1), empty test O(1). Sparse graph (m ~ 4,400 of 87,990 pairs) -> adjacency sets.

3. Technique: why backtracking works

Partial assignments form a tree. The constraint is local: a prefix that breaks one edge can never be repaired, so its entire subtree is cut.

Forward checking: if an unassigned course has no slot left, abandon the branch now.

The cell fixes k (18 slots), so the real question is a **decision** problem: backtracking finds a timetable or proves none exists.

4. Why the other four fail

Technique	Concrete reason
Greedy	Counterexample: a tree needing 2 slots, coordinator's rule uses 3 (slide 8)
Dynamic programming	Substructure only over subsets: 2^n states, O(3^n). Measured 197 ms at n=14, 1.8 s at n=16 (x9 per +2). n=420 impossible
Divide and conquer	Halves are coupled by cross edges; no cheap merge; a half's colouring can make the other half uncolourable
Branch and bound	Fixed k has no objective to bound; for min k the clique bound is weak (Mycielski: clique 2, chi 5 on 23 vertices)

5. Algorithm

```
SEARCH(dom, done, maxc):
  if done == n: return TRUE
  v = unassigned course with fewest allowed slots (tie: most conflicts)
  for c in dom[v]:                      # ascending
      if no pins and c > maxc+1: break  # slot renaming symmetry
      nd = copy(dom); nd[v] = {c}
      for u in adj[v] unassigned: remove c from nd[u]; empty -> reject c
      col[v] = c
      if SEARCH(nd, done+1, max(maxc,c)): return TRUE
      col[v] = unassigned
  return FALSE
```

Min slots: k = clique size, +1 ... until feasible.

6. Trace (Q2): courses in listed order, k = 3

Course	Tried	Result	Reason
DAA	1	accept	
OS	1	REJECT	DAA=1
OS	2	accept	
DBMS	1	REJECT	DAA=1
DBMS	2	REJECT	OS=2
DBMS	3	accept	
ML	1	REJECT	DAA=1
ML	2	accept	
CN	1	accept	OS=2, ML=2
SE	1	REJECT	CN=1
SE	2	accept	
Maths	1	accept	ML=2, SE=2

Slot 1: DAA, CN, Maths | Slot 2: OS, ML, SE | Slot 3: DBMS

Q3: 2 slots are impossible - no search

DAA-OS, DAA-DBMS, OS-DBMS are all conflicts: {DAA, OS, DBMS} is a triangle.

Three pairwise conflicting exams need 3 different slots (pigeonhole). So chi >= 3, and Q2 gives chi = 3.

General rule: a clique of size q forces at least q slots.

Q4: greedy uses more slots than necessary

Tree: u-w, w-z, z-v; leaves l1, l2 on u; m1, m2 on v. Degrees u=v=3, w=z=2, leaves 1. It is 2-colourable: {u,z,m1,m2} | {v,w,l1,l2}.

Order	u	v	w	z	leaves
Slot given	1	1 (not adjacent to u)	2	3 (v=1, w=2)	2

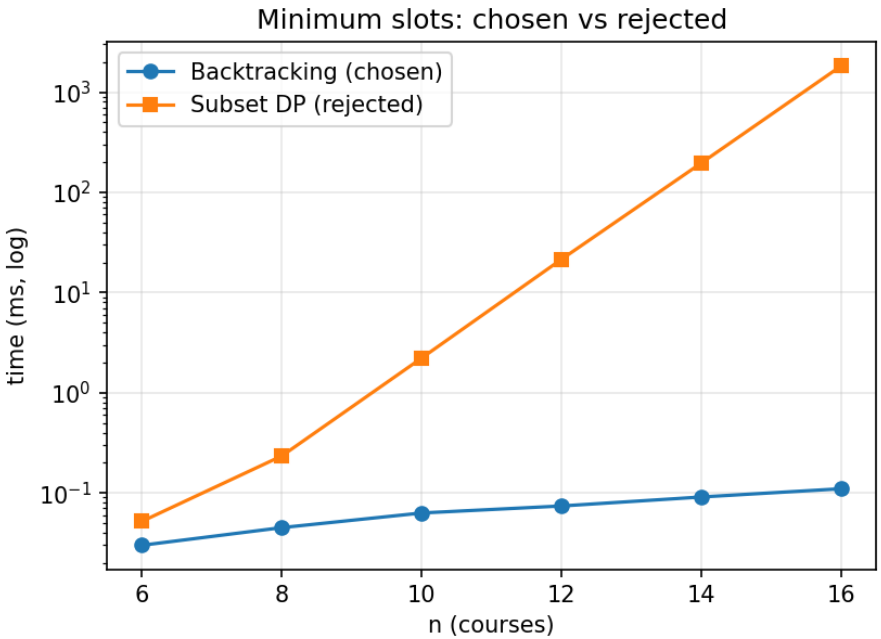
Coordinator: **3 slots**. Optimum: **2**. Greedy never revisits an early choice.

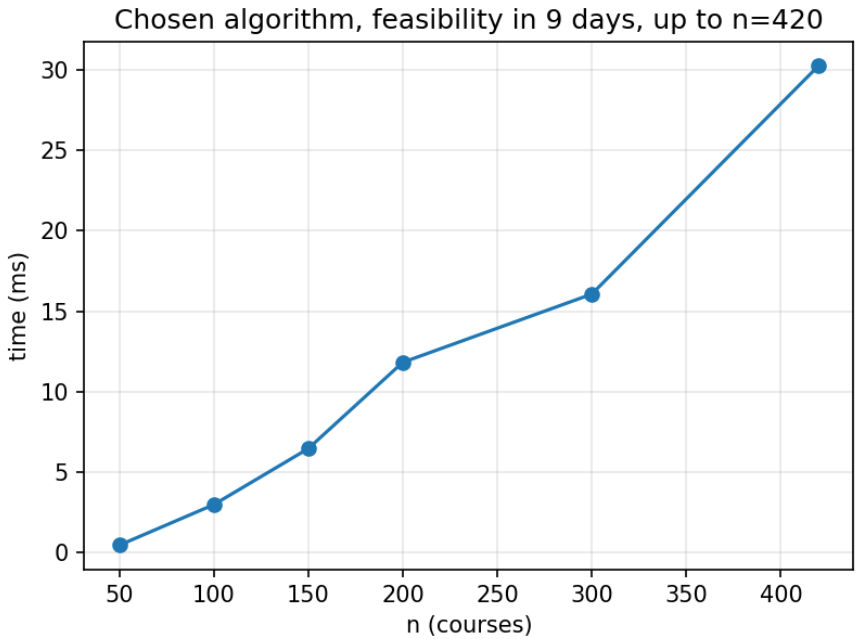
7. Complexity

Case	Time	Why
Worst	O(k^n * n)	T(n) <= k T(n-1) + O(n)
Best	O(n^2 + m)	n nodes, no backtrack
Practical (n=420, k=9)	420 nodes, 30 ms	k >= max conflicts + 1 never fails
Space	O(n^2) as coded, O(n+m) with undo trail	depth n x domain copy

Rejected DP: O(3^n) time, O(2^n) space.

8. Empirical results





DP grows x9 per +2 courses (=3^2). Backtracking at n=420: 30 ms, 420 nodes.

Worst-case input: Mycielski graphs

Triangle-free graphs with chromatic number k, asked for k-1 slots (a 'no' instance that must be fully explored).

chi	n	slots	nodes: most-constrained-first	nodes: input order
4	11	3	26	57
5	23	4	895	49,388
6	47	5	448,614	> 3,000,000 (stopped)

Exponential growth, as predicted. The clique bound (2) is useless: B&B; does not help.

Q5: Maths in slot 1 + one exam per day

Same-day conflict is forbidden -> colour by **day**; slot in the day chosen afterwards. Maths pinned: allowed[Maths] = {D1}.

Best ordering: **most-constrained-first**: Maths (domain size 1), then smallest domain, ties by most conflicts - failures appear near the root.

Step	Pick	Domains left
1	Maths = D1	ML, SE lose D1
2	ML = D2	DAA, CN lose D2
3	DAA = D1	OS, DBMS lose D1
4	OS = D2	DBMS {D3}
5	DBMS = D3	SE {D2}
6-7	SE = D2, CN = D1	done, 0 backtracks

3 slots = 2 days: infeasible. Needs 3 days. Tests (n=40): input order up to millions of nodes, this order always n nodes.

Proof of correctness (sketch)

Invariant: assigned courses never clash along an edge; dom[u] = allowed[u] minus neighbours' slots.

Pruning is safe: an empty domain means no extension exists. Every slot in dom[v] is tried, so a solution below the node is found (induction on n - done). Symmetry cut: unused slots are interchangeable.

Day rule: valid timetable iff conflicting courses are on different days = proper colouring by days; the pin is a unary domain restriction.

Changed requirement and real-world reference

Technique still holds: precolouring + coarser colouring is still a CSP; NP-complete, so exponential worst case remains.

Reference: Brelaz, DSATUR (CACM, 1979); Carter, Laporte and Lee, examination timetabling (JORS, 1996).
Difference: DSATUR is a non-backtracking heuristic; ours adds forward checking and exact proof; real systems add room capacity and soft constraints.